

Resisting Temptation: Is Compulsive Buying an Expression of Personality Deficits?

Laurence Claes^{1,2} · Astrid Müller³

Published online: 7 June 2017
© Springer International Publishing AG 2017

Abstract

Purpose of Review We summarized studies focusing on the associations between compulsive buying and temperament/personality traits in community and clinical samples published between 2010 and 2016.

Recent Findings We found significant associations between compulsive buying (CB) and high levels of Behavioral Activation System (BAS) reactivity, Novelty Seeking, and Positive Urgency (promotion focus), as well as high levels of Neuroticism, Emotional Instability, and Negative Urgency (prevention focus). Finally, the results showed negative relationships between CB and lack of Effortful Control, Self-Control, Perseverance, and Premeditation.

Summary CB seems to be driven by positive reinforcement and negative reinforcement as well as a breakdown in self-regulation. Longitudinal studies should investigate the development of CB throughout time, and experimental studies should examine whether individuals with CB suffer from a general vs. a cue-specific breakdown in self-regulation. Psychotherapy should focus on the training of self-control as well as the development of behavioral alternatives to regulate emotions.

Keywords Compulsive buying · Temperament · Personality · Impulsivity · Big five

Introduction

CB can be defined as a preoccupation with buying, by frequent buying episodes, or urges to buy that are experienced as irresistible [1]. The buying episodes are accompanied by pleasure (positive reinforcement) and relief (negative reinforcement), but followed by remorse and guilt due to psychological, social, occupational and financial problems related to maladaptive spending [2•]. The prevalence of CB in adults is approximately 5% and younger people are more prone to develop CB [3]. Patients with CB suffer from both Axis I (e.g., anxiety, depression, substance abuse, binge eating) and Axis II (e.g., avoidant, obsessive-compulsive and borderline personality disorders (PDs)) comorbid disorders [2•]. Previous research showed that 90% of patients with CB showed at least one life-time Axis I diagnosis and that 60% reported at least one Axis II diagnosis [4]. Recently, several authors reported significantly positive associations between CB and the Cluster B borderline ($r = .42, p < .001$) and narcissistic ($r = .31, p < .001$) [5, 6] personality disorder features. Among many other vulnerability factors (such as materialistic value endorsement, decision-making difficulties, low self-esteem, and depression) [2•], individual differences in personality functioning are considered as possible precursors of CB in both community and clinical samples [7].

The aim of the present review is to offer an overview of studies which focused on the associations between CB and different dimensional temperament/personality models between 2010 and 2016. Findings concerning the associations between CB and temperament/personality traits published before 2010 can be found in a review by

This article is part of the Topical Collection on *Shopping Addiction*

✉ Laurence Claes
Laurence.claes@kuleuven.be

¹ Faculty of Psychology and Educational Sciences, KU Leuven, Tiensestraat 102, box 3720, 3000 Leuven, Belgium

² Faculty of Medicine and Health Sciences (CAPRI), University of Antwerp, Antwerp, Belgium

³ Department of Psychosomatic Medicine and Psychotherapy, Hannover Medical School, Hannover, Germany

Claes and Müller [8••]. In what follows, we discuss the associations between CB and (a) the reactive and regulative temperamental model of Gray and Rothbart [9, 10], (b) the five personality dimensions of the Big Five Model [11], (c) the temperament and character dimensions of the Psychosocial Theory [12], and (d) the impulsivity dimensions of Barratt [13] and the UPPS-P scales [14] (see Table 1).

CB and Reactive and Regulative Temperament

First, we investigated CB from a temperamental perspective. Rothbart, Ahadi and Evans [37] define temperament as individual differences in reactivity and self-regulation. “Reactivity refers to the excitability, responsivity, or arousability of the behavioral and physiological systems of the organism, whereas self-regulation refers to neural and behavioral processes functioning to modulate this underlying reactivity” [37, p. 123]. Reactive temperament (bottom-up) can be conceptualized in terms of two separate neurobiological systems: the Behavioral Inhibition System (BIS) and the Behavioral Activation System (BAS) [9]. The BIS is sensitive to stimuli that signal punishment and omission/termination of reward, and is involved in avoidance behavior. The BIS is related to negative affectivity and personality traits like Neuroticism and Harm Avoidance. The BAS is sensitive to stimuli that signal reward and relief from punishment and is involved in approach behavior. The BAS is related to positive affectivity and personality traits like Extraversion and Sensation Seeking [15]. The reactivity of the BIS/BAS systems can be assessed by means of the BIS/BAS scales [38] or the Sensitivity to Punishment and Sensitivity to Reward Scale (SPSRQ) [39]. Besides reactive temperament, also self-regulation (top-down) can contribute to human behavior. Self-regulation is often synonymously used with self-control [40] or effortful control [10] and is related to personality traits like Conscientiousness, Persistence, and Self-Directedness. Self-regulation may be assessed by means of the Effortful Control Scale of the Adult Temperament Questionnaire measuring Attentional Control, Activation Control and Inhibitory Control [41] or the Self-Control Scale [42]. Studies that have investigated the associations between CB and reactive and regulative temperament in community and clinical samples (see Table 1) showed that CB is significantly related with high levels of BAS reactivity/reward sensitivity and with low levels of effortful control/self-regulation, particularly in females [15, 16, 19–22]. In males, on the contrary, CB seemed to be related to high levels of BIS reactivity, as well as with, as in females, low levels of effortful control [17].

Compulsive Buying and the Big Five Personality Traits

The Five-Factor Model (FFM) organizes personality traits into five higher-order factors being Neuroticism (emotional instability), Extraversion (energetic, outgoing), Openness (creative, open-minded), Agreeableness (friendly/compassionate) and Conscientiousness (well-organized, efficient), and 30 lower-order facets (six for each factor) [11]. To assess the Big Five personality traits and their facets, the most commonly used questionnaires are the NEO-Five Factor Inventory (NEO-FFI, 60 items) and the NEO-Personality Inventory-Revised (NEO-PI-R, 240 items) [11, 43]. However, also shorter questionnaires were developed to assess the Big Five traits such as the Big Five Mini Marker Scale [44] and the mini-International Personality Item Pool (mini-IPIP) [45]. Studies which investigated the association between CB and the Big Five personality traits in community and clinical samples, most often in females, showed that CB is positively related with Neuroticism and negatively with Agreeableness and Conscientiousness [24–27]. Only studies performed in Pakistan [23, 28] did not confirm these findings and showed positive associations between CB and all Big Five personality traits.

CB and the Character and Temperament Dimensions of the Psychosocial Theory of Cloninger

The Psychosocial Theory of Cloninger [12] describes four character dimensions, being Harm Avoidance (anxious, doubtful), Novelty Seeking (thrill-seeking), Reward Dependence (sensitive to social reward) and Persistence (perseverance in spite of fatigue/frustration) and three temperament dimensions, being Self-Directedness (conscientiousness), Cooperativeness (agreeableness), and Self-Transcendence (experiencing ritualistic ideas). The Temperament and Character Inventory-Revised (TCI-R) [46] is frequently used to assess the dimensions of the Psychosocial Theory. Three of the four studies which have investigated the associations between CB and the temperament and character dimensions showed that CB is positively related with Novelty Seeking (i.e., thrill-seeking) in patients with CB, bipolar disorder and substance-use disorders [30–32].

CB and the Impulsivity Dimensions of the Barratt and UPPS-P Scales

Impulsivity may be defined as the tendency to act without prior reflection or thought. The Barratt Impulsiveness Scale-

Table 1 Overview of studies investigating the associations between CB/impulsive buying and temperament/personality dimensions

Authors Country	Sample, gender distribution (%), mean age $\pm$ SD (years), mean baseline BMI $\pm$ SD (year)	Instruments	Analysis	Results
Reactive and Regulative Temperament (Gray, 1982; Rothbart, 1989)				
Claes et al. [15] Belgium	130 undergraduate psychology students 100% females Age: 22.3 (3.6) years	BIS/BAS Scales ATQ-SF-EC + SCS CBS	Correlations	CB correlated significantly positively with BAS-Fun Seeking ($r = .33$), BAS-Drive ($r = .31$), BAS-Reward Responsiveness ($r = .37$) and negatively with Effortful Control (combination of SCS + EC) ($r = -.37$).
Claes et al. [16] Belgium + Germany	292 Belgian and German undergraduate students 72.3% females Age: 22.56 (33.9) years	BIS/BAS Scales ATQ-SF-EC CBS	Correlations	CB correlated significantly positively with BAS-Total ($r = .30$), BAS-Fun Seeking ($r = .33$) BAS-Reward Responsiveness ($r = .25$) and negatively with Effortful Control-Total ($r = -.39$), EC-Inhibitory Control ($r = -.30$), EC-Activation Control ($r = -.34$) and EC-Attentional Control ($r = -.27$).
Mueller et al. [17] Belgium + Germany	410 Belgian and German undergraduate students 70% females Age: 22.9 (3.7) years	BIS/BAS Scales ATQ-SF-EC CBS	Correlations	In male students, CB correlated significantly positively with BIS ($r = .20$) and negatively with EC ($r = -.32$). In female students, CB correlated significantly positively with BAS ($r = .22$) and negatively with EC ($r = -.35$).
Davenport et al. [18] UK	134 students at a university in the East Midlands of the UK 100% females Age: majority between 18 and 24 years	SR BIS-11 CBS	Correlations	CB correlated significantly positively with Sensitivity to Reward ($r = .41$) and Impulsivity ($r = .46$).
Achtziger et al. [19] Germany	Representative sample of the German population ($n = 946$) 55.39% females Age: 48.96 (17.54) years	SCS GABS	Correlations	CB was significantly negatively related with self-control ($r = -.41$).
Claes et al. [20] Belgium	60 patients with an eating disorder 100% females Age: 27.82 (9.76) years	BIS/BAS Scales ATQ-SF-EC DAPP-BQ CBS	Correlations	CB correlated significantly positively with DAPP-Affective Liability ($r = .29$), DAPP-Stimulus Seeking ($r = .30$), and negatively with Effortful Control-Total ($r = -.43$), EC-Inhibitory Control ($r = -.41$), EC-Activation Control ($r = -.43$).
Müller et al. [21] Germany	102 patients with CB 77.5% females Age: 39.2 (11.7) years	BIS/BAS Scales ATQ-SF-EC CBS	Correlations	In patients with CB, CB correlated significantly positively with BAS ($r = .28$).
Voth et al. [22]	31 patients with CB 28 matched healthy controls	BIS/BAS Scales ATQ-SF-EC CBS	ANOVA	CB patients scored significantly higher on BIS, BAS and EC compared to healthy controls. After controlling for

Table 1 (continued)

Authors Country	Sample, gender distribution (%), mean age $\pm$ SD (years), mean baseline BMI $\pm$ SD (year)	Instruments	Analysis	Results
Big Five Model (Costa & McCrae, 1992)	80.6% females (CB) 82.1% females (HC) Age _{CB} : 40.65 (11.26) Age _{HC} : 39.07 (10.29)			depression, only the difference on BAS reactivity remained significant.
Shahjehan et al. [23] Pakistan	640 students of a public sector university of Pakistan 42.8% females Age: between 20y–30y (95.5%)	Big Five Mini Markers Compulsive Buying (Valence et al., 1988)	Correlations	CB was positively related to Neuroticism ($r = .22$), Extraversion ($r = .22$), Openness ($r = .17$), Agreeableness ($r = .18$), and Conscientiousness ($r = .19$).
Donnelly et al. [24] USA	1077 students (sample 1) 74% females Age: 24.23 (9.71) years 650 students (sample 2) 67.7% females Age: 30.27 (12.91) years 667 adults general Galician population 100% females Age: 38.4 (6.1)	Big Five Mini Markers CBS NEO-PI-R GABS	Correlations Correlations	In sample 1/2, CB correlated significantly positively with Neuroticism ($r = .25/.19$) and negatively with Agreeableness ($r = -.22/-.12$), Conscientiousness ($r = -.19/-.23$) and Openness ($r = -.11/ns$). CB was significantly positively related to Neuroticism ($r = .33$) and negatively related to Agreeableness ($r = -.17$) and Conscientiousness ($r = -.27$).
Otero-Lopez et al. [25] Spain	1365 adults general Galician population 51.6% females Age: 39.7 (13.9)	NEO-PI-R GABS	MANCOVA	The high CB group scored significantly higher on Neuroticism (facets) and lower on Conscientiousness (facets) and Agreeableness (facets) compared to the low and moderate CB groups. The high CB group scored also significantly higher on the Extraversion facet of excitement-seeking and lower on the Extraversion facets of Positive Emotions and Assertiveness.
Andreassen et al. [27] Norway	218 undergraduate students 792% females Age: 20.6 (3.0)	NEO-FFI-R CBS	Correlations	CB correlated significantly positively with Neuroticism ($r = .17$) and negatively with Agreeableness ($r = -.24$) and Conscientiousness ($r = -.28$).
Cheema et al. [28] Pakistan	400 individuals of the city of Karachi (Pakistan) 50% females Age: > 18 years	Mini-IPIP RCBS	Correlations	CB was significantly positively related to Neuroticism ($r = .96$), Extraversion ($r = .83$), Intellect ($r = .96$), Agreeableness ($r = .88$), and Conscientiousness ($r = .96$).
Mueller et al. [29]	68 patients with CB 85% females Age: 40.6 (10.6) years	NEO-FFI BIS-11 CBS	K-means cluster analysis	Cluster I of CB patients was characterized by positive z-scores on each of the Big Five personality traits; whereas Cluster II was characterized by a positive

Table 1 (continued)

Authors Country	Sample, gender distribution (%), mean age $\pm$ SD (years), mean baseline BMI $\pm$ SD (year)	Instruments	Analysis	Results
Psychosocial model of Cloninger (1999)				
Black et al. [30] USA	26 persons with CB 32 healthy controls (HC) 88% females CB 84% females controls Age _{CB} : 36.6 (15.7) Age _{HC} : 39.7 (14.8) 158 patients with Bipolar Disorder 59% females Age _{BD} : 48.7 (12.7) years	TCI BIS-11 CBS	ANOVA	z-score on Neuroticism and negative z-scores on Extraversion, Openness, Agreeableness and Conscientiousness. CB patients of Cluster II scored significantly higher on the Barratt Impulsiveness Scale than CB patients of Cluster I. CB scored significantly higher on TCI-Novely Seeking and on BIS-Total, BIS-Attentional, BIS-Motor, BIS-Non-Planning Impulsivity compared to healthy controls.
Di Nicola et al. [31] Italy	158 patients with Bipolar Disorder 59% females Age _{BD} : 48.7 (12.7) years	TCI-R BIS-11 CBS	Correlations	CB correlated significantly positively with TCI-Novely Seeking ($r = .30$), Self-Transcendence ($r = .25$) and BIS-11 Impulsivity ($r = .35$); and negatively with TCI-Self-Directedness ($r = -.28$) and Cooperativeness ($r = -.30$).
Di Nicola et al. [32] Italy	95 Alcohol Use Disorder patients 140 healthy controls 40% females (AUD) 39.3% females (HC) Age _{AUD} : 44.1 (9.5) years Age _{HC} : 42.4 (12.2) years	TCI-R BIS-11 CBS	Correlations	In AUD patients, CB correlated significantly positively with TCI-Novely Seeking ($r = .33$) and BIS-11 Impulsivity ($r = .24$). In healthy controls, CB correlated significantly positively with TCI-Novely Seeking ($r = .38$) and BIS-11 Impulsivity ($r = .30$).
Granero et al. [33] Spain	110 patients with CB disorder 67.3% females Cluster 1: male CB ($n = 32$) Cluster 2: female low dysfunctional ($n = 32$) Cluster 3: female high dysfunctional ($n = 46$) Age: 432 (?) years	TCI-R	Agglomerative hierarchical clustering method	The 'male CB' reported the lowest levels of TCI-Reward Dependence. The 'female low-dysfunctional CB' scored the lowest on TCI-Harm Avoidance and the highest on TCI-Persistence, Self-Directedness, and Cooperativeness. The 'female high dysfunctional CB' reported the highest levels of TCI-Harm Avoidance and the lowest levels of TCI-Self-Directedness.
Impulsivity traits				

Table 1 (continued)

Authors Country	Sample, gender distribution (%), mean age $\pm$ SD (years), mean baseline BMI $\pm$ SD (year)	Instruments	Analysis	Results
Maraz et al. [34] Hungary	1409 Hungarian shopping mall customers 63% females Age: 31.2 (12.1) years 49 compulsive buyers	BIS-11 ECBS	Correlations	CB correlated significantly positively with BIS-11 Impulsivity ($r = .39$).
Williams et al. [35]	37 healthy controls % females:? Age _{CB} : 29.02 Age _{HC} : 25.61 (7.38) years (11.24) years	UPPS-P CBS RCBS	Correlations	CB correlated significantly positive with Positive Urgency ($r = .38/.41$), Negative Urgency ($r = .50/.55$), Lack of Premeditation ($r = .41/.31$) and Lack of Perseverance ($r = .39/.31$) as measured with CBS/RCBS.
Rose et al. [36] USA	514 North American Adults 54% female 32.97 (11.18) years	Mini-IPIP UPPS-P RCBS	Correlations	CB correlated significantly positively with IPIP-Neuroticism ($r = .17$), Negative Urgency ($r = .43$) and Positive Urgency ($r = .41$).

Measures of compulsive buying: *CBS* Compulsive Buying Scale [51], *ECBS* Edwards Compulsive Buying Scale [52], *GABS* German Addictive Buying Scale [53], *RCBS* Richmond Compulsive Buying Scale [54]; Compulsive Buying [55]; Measures of reactive and regulative temperament: *BIS/BAS* Behavioral Inhibition System and Behavioral Activation System Scales [38], *SPSRQ* Sensitivity to Punishment and Sensitivity to Reward Questionnaire [39], *ATQ-SF-EC* Adult Temperament Questionnaire-Short Form-Effortful Control Scale [41], *SCS* Self-Control Scale [42]; Measures of Big Five personality traits: *DAPP-BQ* Dimensional Assessment of Personality Pathology-Basic Questionnaire [56], *mini-IPIP* mini-International Personality Item Pool—Five-Factor Model [45]; Big Five Mini Markers [44]; *NEO-FFI/NEO-PI-R* NEO-Five Factor Inventory (60 items)/NEO Personality Inventory-Revised (240 items) [11]; *NEO-FFI-R* NEO-Five Factor Inventory-Revised [43]; Measures of Psychosocial model: *TCI-R* Temperament and Character Inventory-(Revised) [12, 46]; Measures of Impulsivity traits: *BIS-11* Barratt Impulsiveness Scale-11 [47]; *UPPS-P* UPPS-P Impulsivity Scale [14, 48].

11 (BIS-11) [13, 47] is often used to assess impulsivity. The BIS-11 has been found to factor into three subscales including attentional impulsivity (not paying attention), motor impulsivity (acting without thinking) and non-planning impulsivity (opposite of planning carefully) [47]. Many studies have investigated the associations between CB and BIS-11 (sub)scales and shown positive associations between CB and (attentional, motor and non-planning) impulsiveness in both community and clinical samples [29–32, 34].

Several studies which investigated the association between CB and impulsivity made use of the UPPS-P model (including five impulsivity features of the FFM) [14, 48]. The UPPS-P model consists of the following five dimensions: Negative Urgency (acting impulsively in the presence of negative affect), Lack of Perseverance (not easily stopping an action in progress), Lack of Premeditation (acting without thinking), Sensation Seeking, and Positive Urgency (acting impulsively in the presence of positive affect). The UPPS-P model can be assessed by means of the UPPS-P Impulsivity Scale [48]. Both studies which investigated the associations between CB and Negative/Positive Urgency showed positive associations between CB and negative/positive urgency [35, 36]. Also, two other UPPS-P dimensions, Lack of Perseveration and Lack of Premeditation, showed positive associations with CB [35].

Conclusions

Based on the findings of this review, we conclude that CB is significantly related to both reactive (bottom-up) and regulative (top-down) temperament features. First, CB is significantly related to Neuroticism, Behavioral Inhibition System Reactivity (in males), Affective Lability and Negative Urgency which can be related to the negative reinforcement function of CB. CB can thus be considered as a mean to decrease negative affect (prevention focus) [7], which is consistent with findings from a study using ecological momentary assessment (EMA) [49]. Second, CB is also related to Behavioral Activation System reactivity, Stimulus Seeking, Novelty Seeking, and Positive Urgency, which can be related to the positive reinforcement function of CB. CB can be considered as a mean to gain pleasure, a “kick” or a thrill, and a feeling of being rewarded (i.e., promotion focus) [7]. In sum, from a reactive temperament point of view, CB may be viewed as a way to gain pleasure/reward (promotion focus) and to avoid negative affect or punishment (prevention focus). Longitudinal research would help address the question of whether the functionality of CB changes throughout time (e.g., from positive to negative reinforcement). Does CB start as a mean to acquire positive affect, and

later on (when CB becomes problematic or addictive) as a mean to decrease negative affect? Or do there exist different types of individuals with CB, of whom some use repetitive shopping/buying as a means to gain reward and others use it to avoid negative affect? Finally, almost all studies show that CB is related to deficits in top-down or effortful control, self-control, and self-directedness [7]. Future studies should investigate whether individuals with CB show a general deficit in self-regulation, or whether their effortful control is particularly vulnerable in the presence of “favorite buying items” [50]. Given that many patients with CB have predominantly shifted to online shopping, it is of growing interest to investigate if the aforementioned pattern differs between traditional CB (i.e., buying in “bricks-and-mortar-based” stores) and CB via the Internet. Based on these findings, cognitive behavioral therapy (CBT) may be the treatment of choice for patients with CB given that CBT offers stimulus control techniques, exposure and response prevention which may help to increase effortful control, as well as the development of alternative behaviors (emotion regulation strategies) to gain reward and to reduce negative affect which may often drive CB [2••].

Compliance with Ethical Standards

Conflict of Interest Dr. Laurence Claes and Dr. Astrid Müller declare that they have no conflicts of interest.

Human and Animal Rights and Informed Consent This article does not contain any studies with human or animal subjects performed by any of the authors.

References

Papers of particular interest, published recently, have been highlighted as:

•• Of major importance

1. McElroy SL, Keck PE, Pope HG, Smith JM, Strakowski SM. Compulsive buying: a report of 20 cases. *J Clin Psychiatry*. 1994;55:242–8.
2. Müller A, Mitchell JE, de Zwaan M. Compulsive buying. *Am J Addict*. 2015;24:132–7. **This manuscript contains state-of-the-art information regarding compulsive buying disorder.**
3. Maraz A, Griffiths MD, Demetrovics Z. The prevalence of compulsive buying: a meta-analysis. *Addiction*. 2016;111:408–19. doi: [10.1016/j.addbeh.2016.04.003](https://doi.org/10.1016/j.addbeh.2016.04.003).
4. Mueller A, Mitchell JE, Black DW, Crosby RD, Berg K, de Zwaan M. Latent profile analysis and comorbidity in a sample of individuals with compulsive buying disorder. *Psychiatry res*. 2010;178: 348–53. doi:[10.1016/j.psychres.2010.04.021](https://doi.org/10.1016/j.psychres.2010.04.021).
5. Sansone RA, Chang J, Jewell B, Sellbom M, Bidwell M. Compulsive buying and borderline personality symptomatology. *J Personal Disord*. 2013;27:260–8.

6. Harnish RJ, Bridges KR. Compulsive buying: the role of irrational beliefs, materialism, and narcissism. *J Ration Emot Cogn Behav Ther*. 2015;33:1–16. doi:[10.1007/s10942-014-0197-0](https://doi.org/10.1007/s10942-014-0197-0).
7. Verplanken B, Sato A. The psychology of impulsive buying: an integrative self-regulation approach. *J Consum Policy*. 2011;34:197–210. doi:[10.1007/s10603-011-9158-5](https://doi.org/10.1007/s10603-011-9158-5).
8. Claes L, Mueller A. Personality and compulsive buying disorder. In: Mueller A, Mitchell JE, editors. *Compulsive buying: clinical foundations and treatment*. New York: Routledge Taylor & Francis Group; 2010. **This book chapter reviewed the literature concerning the association between personality and compulsive buying disorder through 2010. Compulsive buying is related to neuroticism (anxiety, negative affectivity) and diminished self-regulation, effortful control or conscientiousness.**
9. Gray JA. *The neuropsychology of anxiety: an inquiry into the functions of the septal-hippocampal system*. Oxford: Oxford University Press; 1982.
10. Rothbart MK. Temperament and development. In: Kohnstamm GA, Bates JE, Rothbart MK, editors. *Temperament in childhood*. Oxford: John Wiley; 1989.
11. Costa PT, McCrae RR. NEO-PI-R: the revised NEO personality inventory. Odessa: Psychological Assessment Resources; 1992.
12. Cloninger CR, Svrakic DM, Przybeck TR. A psychobiological model of temperament and character. *Arch Gen Psychiatr*. 1993;50:975–90.
13. Barratt ES. Impulsiveness substrates: arousal and information processing. In: Spence JT, Izard CE, editors. *Motivation, emotion, and personality*. Elsevier Science Publishers: North Holland; 1985.
14. Cyders MA, Smith GT, Spillane NS, Fischer S, Annus A, Peterson C. Integration of impulsivity and positive mood to predict risky behavior: development and validation of a measure of positive urgency. *Psychol Assess*. 2007;19:107–18. doi:[10.1037/1040-3590.19.1.107](https://doi.org/10.1037/1040-3590.19.1.107).
15. Claes L, Bijttebier P, Van den Eynde F, Mitchell JE, Faber R, de Zwaan M, et al. Emotional reactivity and self-regulation in relation to compulsive buying. *Personal Individ Differ*. 2010;49:526–30. doi:[10.1016/j.paid.2010.05.020](https://doi.org/10.1016/j.paid.2010.05.020).
16. Claes L, Bijttebier P, Mitchell JE, de Zwaan M, Mueller A. The relationship between compulsive buying, eating disorder symptoms, and temperament in a sample of female students. *Compr Psychiatry*. 2011;52:50–5. doi:[10.1016/j.comppsy.2010.05.003](https://doi.org/10.1016/j.comppsy.2010.05.003).
17. Mueller A, Claes L, Mitchell JE, Faber RJ, Fischer J, de Zwaan M. Does compulsive buying differ between male and female students? *Personal Individ Differ*. 2011;50:1300–12. doi:[10.1016/j.paid.2011.02.026](https://doi.org/10.1016/j.paid.2011.02.026).
18. Davenport K, Houston JE, Griffiths MD. Excessive eating and compulsive buying behaviours in women: an empirical pilot study examining reward sensitivity, anxiety, impulsivity, self-esteem and social desirability. *Int J Ment Health Addict*. 2012;10:474–89. doi:[10.1007/s11469-011-9332-7](https://doi.org/10.1007/s11469-011-9332-7).
19. Achtziger A, Hubert M, Kenning P, Raab G, Reisch L. Debt out of control: the links between self-control, compulsive buying, and real debts. *J Econ Psychol*. 2015;49:141–9. doi:[10.1016/j.joep.2015.04.003](https://doi.org/10.1016/j.joep.2015.04.003).
20. Claes L, Müller A, Norré J, Van Assche L, Wonderlich S, Mitchell JE. The relationship among compulsive buying, compulsive internet use and temperament in a sample of female patients with eating disorders. *Eur Eat Disord Rev*. 2012;20:126–31. doi:[10.1002/erv.1136](https://doi.org/10.1002/erv.1136).
21. Müller A, Claes L, Georgiadou E, Möllenkamp M, Voth EM, Faber RJ, et al. Is compulsive buying related to materialism, depression, or temperament? Findings from a sample of treatment-seeking patients with CB. *Psychiatry Res*. 2014;216:103–7. doi:[10.1016/j.psychres.2014.01.012](https://doi.org/10.1016/j.psychres.2014.01.012).
22. Voth EM, Claes L, Georgiadou E, Selle J, Trotske P, Brand M, et al. Reactive and regulative temperament in patients with compulsive buying and non-clinical controls measured by self-report and performance-based tasks. *Compr Psychiatry*. 2014;55:1505–12. doi:[10.1016/j.comppsy.2014.05.011](https://doi.org/10.1016/j.comppsy.2014.05.011).
23. Shahjehan A, Qureshi JA, Zeb F, Saifullah K. The effect of personality on impulsive and compulsive buying behaviors. *Afr J Bus Manag*. 2012;6:2187–94.
24. Donnelly G, Ksendzova M, Howell RT. Sadness, identity, and plastic in over-shopping: the interplay of materialism, poor credit management, and emotional buying motives in predicting compulsive buying. *J Econ Psychol*. 2013;39:113–25. doi:[10.1016/j.joep.2013.07.006](https://doi.org/10.1016/j.joep.2013.07.006).
25. Otero-Lopez JM, Villardefrancos E. Five-factor model personality traits, materialism, and excessive buying: a mediational analysis. *Personal Individ Differ*. 2013;54:767–72. doi:[10.1016/j.paid.2012.12.013](https://doi.org/10.1016/j.paid.2012.12.013).
26. Otero-Lopez JM, Villardefrancos-Pol E. Compulsive buying and the five-factor model of personality: a facet analysis. *Personal Individ Differ*. 2013;55:585–90. doi:[10.1016/j.paid.2013.05.005](https://doi.org/10.1016/j.paid.2013.05.005).
27. Andreassen CS, Griffiths MD, Gjertsen SR, Krossbakken E, Kvam S, Pallesen S. The relationships between behavioral addictions and the five-factor model of personality. *J Behav Addict*. 2013;2:90–9. doi:[10.1556/JBA.2.2013.003](https://doi.org/10.1556/JBA.2.2013.003).
28. Cheema F, Phanwar I, Zia S, Rasool M. Exploring the personality traits as cause of compulsive buying behavior. *J Bus Strateg*. 2014;8:19–29.
29. Mueller A, Claes L, Mitchell JE, Wonderlich SA, Crosby RD, de Zwaan M. Personality prototypes in individuals with compulsive buying based on the big five model. *Behav Res Ther*. 2010;48:930–5. doi:[10.1016/j.brat.2010.05.020](https://doi.org/10.1016/j.brat.2010.05.020).
30. Black DW, Shaw M, McCormick B, Bayless JD, Allen J. Neuropsychological performance, impulsivity, ADHD symptoms, and novelty seeking in compulsive buying disorder. *Psychiatry Res*. 2012;200:581–7. doi:[10.1016/j.psychres.2012.06.003](https://doi.org/10.1016/j.psychres.2012.06.003).
31. Di Nicola M, Tedeschi D, Mazza M, Martinotti G, Harnic D, Catalano V, et al. Behavioural addictions in bipolar disorder patients: role of impulsivity and personality dimensions. *J Affect Disord*. 2010;125:82–8. doi:[10.1016/j.jad.2009.12.016](https://doi.org/10.1016/j.jad.2009.12.016).
32. Di Nicola M, Tedeschi D, De Risio L, Pettorruso M, Martinotti G, Ruggeri F, et al. Co-occurrence of alcohol use disorder and behavioural addictions: relevance of impulsivity and craving. *Drug Alcohol Depend*. 2015;148:118–25. doi:[10.1016/j.drugalcdep.2014.12.028](https://doi.org/10.1016/j.drugalcdep.2014.12.028).
33. Granero R, Fernandez-Aranda F, Bano M, Steward T, Mestre-Bach G, del Pino-Gutierrez A, et al. Compulsive buying disorder clustering based on sex, age, onset and personality traits. *Compr Psychiatry*. 2016;68:1–10. doi:[10.1016/j.comppsy.2016.03.003](https://doi.org/10.1016/j.comppsy.2016.03.003).
34. Maraz A, Urban R, Demetrovics Z. Borderline personality disorder and compulsive buying: a multivariate etiological model. *Addict Behav*. 2016;60:117–23. doi:[10.1016/j.addbeh.2016.04.003](https://doi.org/10.1016/j.addbeh.2016.04.003).
35. Williams AD, Grisham JR. Impulsivity, emotion regulation, and mindful attentional focus in compulsive buying. *Cogn Ther Res*. 2012;36:451–7. doi:[10.1007/s10608-011-9384-9](https://doi.org/10.1007/s10608-011-9384-9).
36. Rose P, Segrist DJ. Negative and positive urgency may both be risk factors for compulsive buying. *J Behav Addict*. 2014;3:128–32. doi:[10.1556/JBA.3.2014.011](https://doi.org/10.1556/JBA.3.2014.011).
37. Rothbart MK, Ahadi SA, Evans DE. Temperament and personality: origins and outcomes. *J Pers Soc Psychol*. 2000;78:122–35. doi:[10.1037/0022-3514.78.1.122](https://doi.org/10.1037/0022-3514.78.1.122).
38. Carver CS, White TL. Behavioural inhibition, behavioural activation, and affective responses to impending reward and punishment: the BIS/BAS scales. *J Pers Soc Psychol*. 1994;67:319–33.
39. Torrubia R, Avila C, Molt OJ, Caseras X. The sensitivity to punishment and sensitivity to reward questionnaire (SPSRQ) as a measure of gray's anxiety and impulsivity dimensions. *Personal Individ Differ*. 2001;31:837–62.

40. Baumeister RF, Heatherton TF, Tice DM. *Losing control: how and why people fail at self-regulation*. San Diego: Academic Press; 1994.
41. Evans DE, Rothbart MK. Developing a model for adult temperament. *J Res Pers*. 2007;41:868–88. doi:[10.1016/j.jrp.2006.11.002](https://doi.org/10.1016/j.jrp.2006.11.002).
42. Tangney JP, Baumeister RF, Boone AL. High self-control predicts good adjustment, less pathology, better grades, and interpersonal success. *J Pers*. 2004;72:271–324. doi:[10.1111/j.0022-3506.2004.00263.x](https://doi.org/10.1111/j.0022-3506.2004.00263.x).
43. McCrae RR, Costa PT. A contemplated revision of the NEO five-factor inventory. *Personal Individ Differ*. 2004;36:587–96. doi:[10.1016/S0191-8869\(03\)00118-1](https://doi.org/10.1016/S0191-8869(03)00118-1).
44. Saucier G. Mini-markers: a brief version of Goldberg's unipolar big-five markers. *J Pers Assess*. 1994;63:506–16.
45. Donnellan MB, Oswald FL, Baird BM, Lucas RE. The mini-IPIP: tiny-yet-effective measures of the big five factors of personality. *Psychol Assess*. 2006;18:192–203. doi:[10.1037/1040-3590.18.2.192](https://doi.org/10.1037/1040-3590.18.2.192).
46. Cloninger CR. *The temperament and character inventory-revised*. St Louis: Center for Psychobiology of Personality; 1999.
47. Patton JH, Stanford MS, Barratt ES. Factor structure of the Barratt impulsiveness scale. *J Clin Psychol*. 1995;51:768–74.
48. Lynam D, Smith GT, Cyders MA, Fischer S, Whiteside SA. *The UPPS-P: A multidimensional measure of risk for impulsive behavior*. Unpublished technical report; 2007.
49. Mueller A, Mitchell JE, Crosby RD, Cao L, Claes L, de Zwaan M. Mood states preceding and following compulsive buying episodes: an ecological momentary assessment study. *Psychiatry Res*. 2012;200:575–80. doi:[10.1016/j.psychres.2012.04.015](https://doi.org/10.1016/j.psychres.2012.04.015).
50. Trotzke P, Stracke K, Müller A, Brand M. Pathological buying online as a specific form of internet addiction: a model-based experimental investigation. *PlosONE*. 2015; doi:[10.1371/journal.pone.0140296](https://doi.org/10.1371/journal.pone.0140296).
51. Faber RJ, O'Guinn TC. A clinical screener for compulsive buying. *J Consum Res*. 1992;19:459–69.
52. Edwards EA. Development of a new scale for measuring compulsive buying behavior financial. *Couns Plan*. 1993;4:67–84.
53. Raab G, Neuner M, Reisch LA, Scherhorn G. *SKSK – Screeningverfahren zur Erhebung von kompensatorischem und süchtigem Kaufverhalten*. Göttingen: Hogrefe; 2005.
54. Ridgway NM, Kukar-Kinney M, Monroe KB. An expanded conceptualization and a new measure of compulsive buying. *J Consum Res*. 2008;35:622–39. doi:[10.1086/591108](https://doi.org/10.1086/591108).
55. Valence G, d'Astous A, Fortier L. Compulsive buying: concept and measurement. *J Consum Policy*. 1988;11:419–33.
56. Livesley WJ, Jackson DN. *Manual for the dimensional assessment of personality pathology – basic questionnaire (DAPP-BQ)*. Port Huron: Sigma Press; 2002.

Terms and Conditions

Springer Nature journal content, brought to you courtesy of Springer Nature Customer Service Center GmbH (“Springer Nature”).

Springer Nature supports a reasonable amount of sharing of research papers by authors, subscribers and authorised users (“Users”), for small-scale personal, non-commercial use provided that all copyright, trade and service marks and other proprietary notices are maintained. By accessing, sharing, receiving or otherwise using the Springer Nature journal content you agree to these terms of use (“Terms”). For these purposes, Springer Nature considers academic use (by researchers and students) to be non-commercial.

These Terms are supplementary and will apply in addition to any applicable website terms and conditions, a relevant site licence or a personal subscription. These Terms will prevail over any conflict or ambiguity with regards to the relevant terms, a site licence or a personal subscription (to the extent of the conflict or ambiguity only). For Creative Commons-licensed articles, the terms of the Creative Commons license used will apply.

We collect and use personal data to provide access to the Springer Nature journal content. We may also use these personal data internally within ResearchGate and Springer Nature and as agreed share it, in an anonymised way, for purposes of tracking, analysis and reporting. We will not otherwise disclose your personal data outside the ResearchGate or the Springer Nature group of companies unless we have your permission as detailed in the Privacy Policy.

While Users may use the Springer Nature journal content for small scale, personal non-commercial use, it is important to note that Users may not:

1. use such content for the purpose of providing other users with access on a regular or large scale basis or as a means to circumvent access control;
2. use such content where to do so would be considered a criminal or statutory offence in any jurisdiction, or gives rise to civil liability, or is otherwise unlawful;
3. falsely or misleadingly imply or suggest endorsement, approval, sponsorship, or association unless explicitly agreed to by Springer Nature in writing;
4. use bots or other automated methods to access the content or redirect messages
5. override any security feature or exclusionary protocol; or
6. share the content in order to create substitute for Springer Nature products or services or a systematic database of Springer Nature journal content.

In line with the restriction against commercial use, Springer Nature does not permit the creation of a product or service that creates revenue, royalties, rent or income from our content or its inclusion as part of a paid for service or for other commercial gain. Springer Nature journal content cannot be used for inter-library loans and librarians may not upload Springer Nature journal content on a large scale into their, or any other, institutional repository.

These terms of use are reviewed regularly and may be amended at any time. Springer Nature is not obligated to publish any information or content on this website and may remove it or features or functionality at our sole discretion, at any time with or without notice. Springer Nature may revoke this licence to you at any time and remove access to any copies of the Springer Nature journal content which have been saved.

To the fullest extent permitted by law, Springer Nature makes no warranties, representations or guarantees to Users, either express or implied with respect to the Springer nature journal content and all parties disclaim and waive any implied warranties or warranties imposed by law, including merchantability or fitness for any particular purpose.

Please note that these rights do not automatically extend to content, data or other material published by Springer Nature that may be licensed from third parties.

If you would like to use or distribute our Springer Nature journal content to a wider audience or on a regular basis or in any other manner not expressly permitted by these Terms, please contact Springer Nature at

onlineservice@springernature.com